

MK110 Plus Series BLE Gateway Mini

Product Specification

Version V1.0

Contents

1. Product Introduction 1
    1.1 Overview 1
    1.2 Product Model 1
2. Features 2
3. Appearance 2
    3.1 Product Appearance 2
    3.2 Product Dimensions 3
4. Specification 3
5. User Guide 4
    5.1 How to Install the Gateway? 4
    5.2 How to Configure the Gateway? 5
6. Main Function 5
    6.1 Pairing mode 5
    6.2 Reconnect Network 5
    6.3 Gateway uplink 5
        6.3.1 Scan BLE Devices 5
        6.3.2 Data Filtering 6
        6.3.3 Date Decoding 6
        6.4.4 Data Upload 6
    6.4 Gateway downlink 6
    6.5 OTA 6
    6.6 Restore to Factory Settings 6
7. Development Document 7
8. Revision History 7

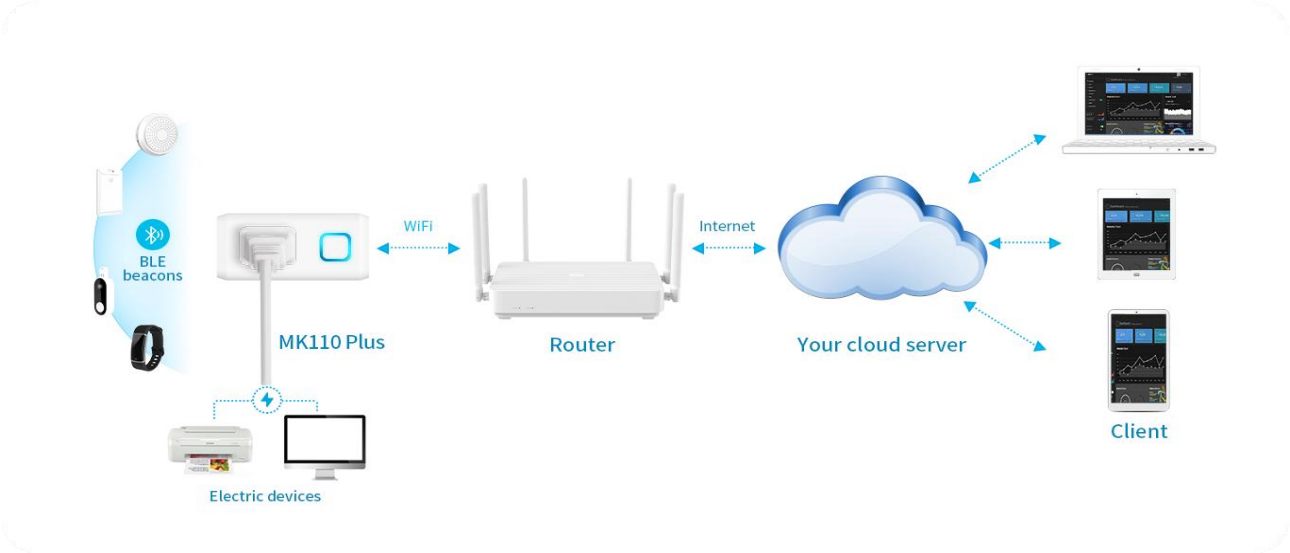
1. Product Introduction

1.1 Overview

MK110 Plus series is a BLE to WIFI gateway, it has a mini body and plug-and-play design, which makes the installation and deployment simple.

MK110 Plus can connect to the customer server through a personal or enterprise (**WPA2-Enterprise**) WIFI network, it also can work with some popular IOT platforms, such as AWS IoT, and Alibaba Cloud IoT.

The gateway is a data bridge between beacon devices and your cloud server, it supports the uplink communication with cloud server, collects beacon advertising data through Bluetooth and uploads them to server through WiFi. It also supports the downlink communication with Beacon devices, users can manage the Beacon device on the cloud by sending commands to the gateway. Based on these features, the gateway can be widely used in real time location service, condition monitoring, assets tracking and personal safety management.



1.2 Product Model

MK110 Plus series includes three product models, their main differences are the WIFI and BLE module, which determine their performance and cost.

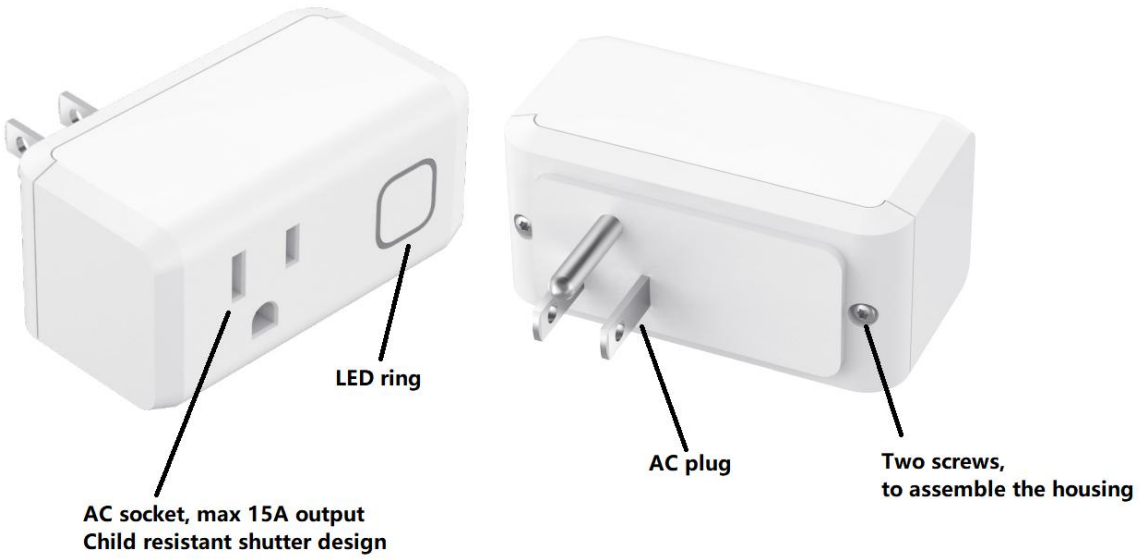
Product model	WIFI module	BLE module	Description
MK110 Plus 01	2.4GHz WIFI and BLE4.2 two-in-one module		/
MK110 Plus 02	2.4GHz WIFI module	BLE 5.0 module	Supports BLE long range
MK110 Plus 03	Dual band WIFI module	BLE 5.0 module	Supports BLE long range

2. Features

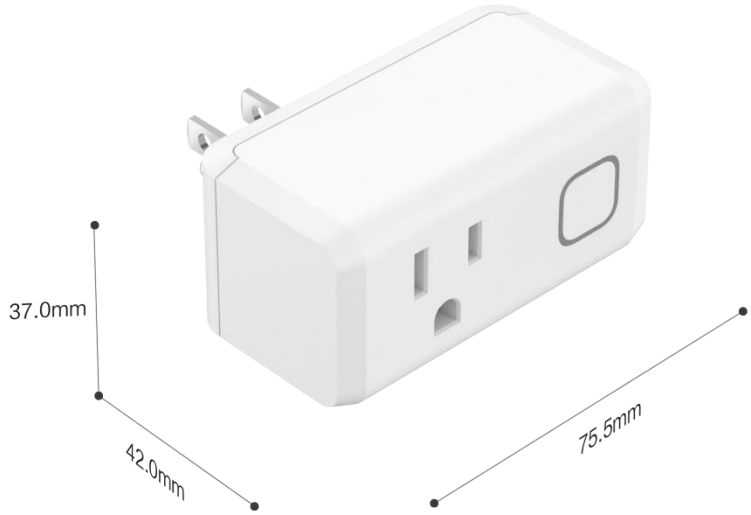
- Mini size, plug and play
- US plug with 15A AC power output
- Child resistant shutter design, designed to inhibit access to the electricity supply
- Supports personal and Enterprise (WPA2-Enterprise) WIFI security, ensures a security network connectivity
- Support connecting to customer's own serve via MQTT, ensuring the data security and privacy
- Real time scanning and continuously uploading data to cloud
- Connect and communicate with the certain Beacon device, users can remotely manage beacon devices on the Cloud
- Efficient and flexible filter settings to help obtain target beacon data, also support filter out the duplicate data, to reduce too many duplicate data uploaded to server
- Locally decode iBeacon, Eddystone(UID/URL/TLM) and all MOKO beacon advertising data
- FCC and UL certified (under process)

3. Product Appearance

3.1 Appearance



3.2 Dimensions



4. Specification

Electronic	
Plug type	US
Power supply	120VAC, 50/60Hz
Output capacity	15A, 1800W max
Button	1* reset button
LED indicator	1* LED indicators
Physical	
Material	ABS + PC
Color	White
Dimension (L*W*H)	75.5mm*42mm*37mm
Environment	
Operating temperature	0 ºC~ 40 ºC
Operating humidity	0%～95% (No condensation)

Storage temperature	-10 ºC~ 50 ºC
Bluetooth	
Protocol	MK110 Plus 01: BLE 4.2 MK110 Plus 02/03: BLE 5.0
Scanning Range	MK110 plus 01: above 100 meters (in an open space)
WIFI	
Band	MK110 Plus 01/02: 2.4GHz MK110 Plus 03: 2.4GHz&5.0 GHz
Protocol	MK110 Plus 01/02: 802.11 b/g/n MK110 Plus 03: 802.11 a/b/g/n
Security	OPEN/WEP/WPA_PSK/WPA2_PSK/WPA_WPA2_PSK/**WPA2-Enterprise**
Communication	
Protocol	MQTT V3.1.1
Encryption	SSL/TLS
Data Format	JSON

5. User Guide

5.1 How to Install the Gateway?

Just plug the gateway into a power socket, the gateway will start work immediately.



5.2 How to Configure the Gateway?

Scan the QR code below to download the MKScannerPro APP. You can also search for the app and download it from Google play or APP store. For the detailed configuration steps, please refer to the ***User Manual*** document.

6. Main Function

6.1 Pairing mode

Paring mode is used to configure WIFI and MQTT settings for the gateway. If the gateway is configured for the first time, after powered up, it will enter paring mode and Bluetooth advertises with name of MK110 Plus -XXXX (XXXX is the last 2 bytes of the MAC address).

The gateway Bluetooth can be scanned and connected by the MOKO APP. After the APP and gateway Bluetooth are successfully connected, users can send WIFI and server information to the gateway via the APP, and then the gateway will automatically connect to the certain WIFI and MQTT server.

6.2 Reconnect Network

If the gateway disconnects from server due to the network issue or power losing, after the network or power supply come back, the gateway will automatically reconnect to the WIFI and server which the gateway is successfully connected before disconnection.

6.3 Gateway uplink

Gateway scans for nearby BLE devices and upload the BLE data to the cloud.

6.3.1 Scan BLE Devices

The gateway supports to collect iBeacon, Eddytsone and BLE 4.2, BLE 5.0 device data.

6.3.2 Data Filtering

The gateway supports multiple data filtering methods to help you easily obtain target beacon data:

- Filter by RSSI, MAC address, advertising name and advertising raw data.
- Filter out duplicate data, reports only non-duplicate data to your server in a filtering period.

6.3.3 Date Decoding

The gateway can locally decode the scanned Bluetooth advertising data before uploading to server. The users can directly obtain the desirable information on the server.

- Support iBeacon, Eddystone (UID/URL/TLM) raw data decoding.
- Support all MOKO beacon raw data decoding.

6.4.4 Data Upload

The gateway uploads Beacon packets to server through WiFi. The uploading data content is configurable, the users can configure the gateway reporting data to include only what they need, which will effectively save server and network resources.

6.4 Gateway downlink

Gateway connects and communicates with Beacon device. The users can remotely manage Beacon devices by sending command to the gateway from cloud.

The typical application is to use the gateway and MOKO B1/B2 panic button for personal security management solution. In hospital, the staffs wear the panic button, when an emergency happens, the staff presses the SOS button to trigger the alarm message advertising. The gateway transfers the alarm message to cloud, and connect to the panic button to inform that the alarm is received by the management team.

6.5 OTA

The gateway has the ability to upgrade firmware over the air. If MOKOsmart releases a new firmware, MOKOsmart can provide the firmware upgrade file with customers, so that customers can upgrade their gateways by a wireless method.

6.6 Restore to Factory Settings

The gateway is equipped with a button for reset operation. Press the button and hold for 10 seconds in 1 minute after it is powered up, the gateway will be reset and restore factory settings, then it will enter pairing mode.

Users can also reset the gateway by sending MQTT commands from the cloud.

7. Development Document

MOKO provides the following documents for customers to test the gateway and develop their own firmware or platform.

File	Description
MK110 Plus Series Product Specification	This document mainly introduces the gateway and guide users to install and configure the gateway.
MK110 Plus User Manual	This document instructs users how to configure the gateway with MOKO APP.
MK110 Plus Communication Protocol	This document describes the configuration commands and management commands supported on MK110 Plus gateway.
APP SDK	iOS: https://github.com/MKScannerPro/MKScannerPro_iOS.git Android: https://github.com/MKScannerPro/MKScannerPro_Android.git The APP SDK includes the source code of the MKScannerPro APP, customers can quickly develop their own APP with it.

8. Revision History

Revision	Description	Editor	Date
V1.0	Initial version	Weiguifen	2023.5.10

MOKO TECHNOLOGY LTD.

4F, Building2, Guanghui Technology Park,
MinQing Rd, Longhua, Shenzhen, Guangdong, China

Tel: 86-755-23573370-829

sales@mokosmart.com

https://www.mokosmart.com